ASSESSMENT 1
BASIC PROGRAMS

NAME: DIYA S

REGISTER NUMBER: 25BAI0235

COURSE NAME: EXPLORATORY DATA ANALYSIS

SLOT: E2+TE2

PROGRAM 1:

Weather data analysis using pandas

import pandas as pd

```
#DIYA S 25BAI0235
print("DIYA S 25BAI0235")
data = {

"Day": ["Mon", "Tue", "Wed", "Thu", "Fri"],

"Temperature": [30, 33, 29, 35, 31],

"Humidity": [65, 70, 75, 60, 68],

"Rainfall": [0, 2, 5, 0, 1]

}
```

```
df = pd.DataFrame(data)

print("WEATHER DATA")

print(df)

print("\nAverage Temperature:", df["Temperature"].mean())

print("Highest Temperature:", df["Temperature"].max())

print("Lowest Temperature:", df["Temperature"].min())

print("\nRainy Days")
```

```
print(df[df["Rainfall"] > 0])



print("\nDays with Temperature Above 30°C")

print(df[df["Temperature"] > 30])
```

OUTPUT:

```
DIYA S 25BAI0235
WEATHER DATA
   Day  Temperature  Humidity  Rainfall
0  Mon           30        65         0
1  Tue           33        70         2
2  Wed           29        75         5
3  Thu           35        60         0
4  Fri           31        68         1

Average Temperature: 31.6
Highest Temperature: 35
Lowest Temperature: 29

Rainy Days
   Day  Temperature  Humidity  Rainfall
1  Tue           33        70         2
2  Wed           29        75         5
4  Fri           31        68         1

Days with Temperature Above 30°C
   Day  Temperature  Humidity  Rainfall
1  Tue           33        70         2
3  Thu           35        60         0
4  Fri           31        68         1
```

PROBLEM 2:

Student Marks Management System Using Dictionary

```
#DIYA S 25BAI0235
print("DIYA S 25BAI0235")
students = {
    "Anu": 85,
    "Rahul": 92,
    "Priya": 78,
    "Kiran": 88,
    "Meena": 95
}

print("Student Marks")
for name, marks in students.items():
    print(name, ":", marks)

top_student = max(students, key=students.get)
print("\nTop Scorer:", top_student)
print("Marks:", students[top_student])
```

```
average = sum(students.values()) / len(students)
print("\nAverage Marks:", average)


search = input("\nEnter student name: ")


if search in students:
    print(search, "scored", students[search], "marks.")
else:
    print("Student not found.")
```

OUTPUT:

```
DIYA S 25BAI0235
Student Marks
Anu : 85
Rahul : 92
Priya : 78
Kiran : 88
Meena : 95

Top Scorer: Meena
Marks: 95

Average Marks: 87.6

Enter student name: Rahul
Rahul scored 92 marks.
```

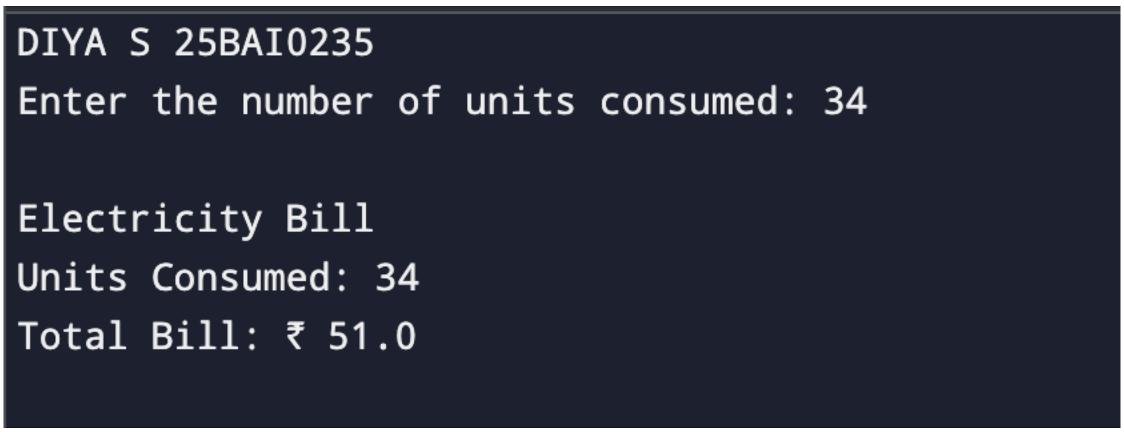
PROBLEM 3:

ELECTRICITY BILL CALCULATOR

print("DIYA S 25BAI0235") units = int(input("Enter the number of units consumed: "))

if units <= 100: bill = units * 1.5 elif units <= 200: bill = (100 * 1.5) + ((units - 100) * 2.5) else: bill = (100 * 1.5) + (100 * 2.5) + ((units - 200) * 4) print("\nElectricity Bill") print("Units Consumed:", units) print("Total Bill: ₹", bill)

OUTPUT:



```
DIYA S 25BAI0235
Enter the number of units consumed: 34

Electricity Bill
Units Consumed: 34
Total Bill: ₹ 51.0
```

PROBLEM 4:

EXPENSE TRACKER

print("DIYA S 25BAI0235")

expenses = {}

```
expenses["Food"] = float(input("Enter Food Expense: "))

expenses["Travel"] = float(input("Enter Travel Expense: "))

expenses["Shopping"] = float(input("Enter Shopping Expense: "))

expenses["Entertainment"] = float(input("Enter Entertainment Expense: "))


print("\nExpense Details")

for category, amount in expenses.items():

    print(category, ":", amount)


total = sum(expenses.values())

print("\nTotal Expense: ₹", total)


highest = max(expenses, key=expenses.get)

print("Highest Expense Category:", highest)

print("Amount: ₹", expenses[highest])
```

OUTPUT:

```
DIYA S 25BAI0235
Enter Food Expense: 4300
Enter Travel Expense: 2300
Enter Shopping Expense: 3444
Enter Entertainment Expense: 2400

Expense Details
Food : 4300.0
Travel : 2300.0
Shopping : 3444.0
Entertainment : 2400.0

Total Expense: ₹ 12444.0
Highest Expense Category: Food
Amount: ₹ 4300.0
```

PROBLEM 5:

BMI CALCULATOR

print("DIYA S 25BAI0235") weight = float(input("Enter your weight (kg): ")) height = float(input("Enter your height (m): "))

bmi = weight / (height ** 2)

print("\nBMI:", round(bmi, 2))

if bmi < 18.5: print("Category: Underweight") elif bmi < 25: print("Category: Normal Weight") elif bmi < 30: print("Category: Overweight") else: print("Category: Obese")

OUTPUT:

```
DIYA S 25BAI0235
Enter your weight (kg): 39
Enter your height (m): 1.5

BMI: 17.33
Category: Underweight
```

PROBLEM 6:

ATM SIMULATION USING EXCEPTION HANDLING

print("DIYA S 25BAI0235")

try:

```
    balance = float(input("Enter Account Balance: "))
    amount = float(input("Enter Withdrawal Amount: "))

    if amount > balance:
        raise ValueError("Insufficient Balance")

    balance = balance - amount

    print("\nWithdrawal Successful")
    print("Remaining Balance: ₹", balance)

except ValueError as e:
    print("Error:", e)
```

OUTPUT:

```
DIYA S 25BAI0235
Enter Account Balance: 23000
Enter Withdrawal Amount: 4000

Withdrawal Successful
Remaining Balance: ₹ 19000.0
```

PROBLEM 7:

Students record file reader

```
print("DIYA S 25BAI0235")
stack = []

while True:
    print("\n1. Push")
    print("2. Pop")
    print("3. Display")
    print("4. Exit")
```

```
choice = int(input("Enter your choice: "))

if choice == 1:
    item = input("Enter element: ")
    stack.append(item)
    print(item, "pushed into stack")

elif choice == 2:
    if len(stack) == 0:
        print("Stack is Empty")
    else:
        print("Popped Element:", stack.pop())

elif choice == 3:
    if len(stack) == 0:
        print("Stack is Empty")
    else:
```

```
print("Stack:", stack)
```

OUTPUT:

```
DIYA S 25BAI0235

1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter element: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 4
Program Ended
```

PROBLEM 8:

Double ended queue

```
print("DIYA S 25BAI0235")
from collections import deque

dq = deque()

while True:
    print("\n1. Insert Front")
    print("2. Insert Rear")
    print("3. Delete Front")
    print("4. Delete Rear")
    print("5. Display")
    print("6. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        item = input("Enter element: ")
```

```
        dq.appendleft(item)
        print(item, "inserted at front")

    elif choice == 2:
        item = input("Enter element: ")
        dq.append(item)
        print(item, "inserted at rear")

    elif choice == 3:
        if len(dq) == 0:
            print("Deque is Empty")
        else:
            print("Deleted:", dq.popleft())

    elif choice == 4:
        if len(dq) == 0:
            print("Deque is Empty")
        else:
            print("Deleted:", dq.pop())
```

```
    elif choice == 5:
        print("Deque:", list(dq))

    elif choice == 6:
        print("Program Ended")
        break

    else:
        print("Invalid Choice")
```

OUTPUT:

```
DIYA S 25BAI0235

1. Insert Front
2. Insert Rear
3. Delete Front
4. Delete Rear
5. Display
6. Exit
Enter your choice: 1
Enter element: 2
2 inserted at front

1. Insert Front
2. Insert Rear
3. Delete Front
4. Delete Rear
5. Display
6. Exit
Enter your choice: 2
Enter element: 2
2 inserted at rear

1. Insert Front
2. Insert Rear
3. Delete Front
4. Delete Rear
5. Display
6. Exit
```

PROBLEM 9:

Queue

queue = []

```
while True:
    print("\n1. Enqueue")
    print("2. Dequeue")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        item = input("Enter element: ")
        queue.append(item)
        print(item, "inserted into queue")

    elif choice == 2:
        if len(queue) == 0:
            print("Queue is Empty")
        else:
            print("Deleted Element:", queue.pop(0))
```

```
    elif choice == 3:
        if len(queue) == 0:
            print("Queue is Empty")
        else:
            print("Queue:", queue)

    elif choice == 4:
        print("Program Ended")
        break

    else:
        print("Invalid Choice")
```

OUTPUT:

```
DIYA S 25bai0235

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 2
2 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
Deleted Element: 2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue is Empty

1. Enqueue
2. Dequeue
3. Display
4. Exit
```

PROBLEM 10:

PALINDROME CHECKER

print("DIYA S 25bai0235")

text = input("Enter a string: ")

```
reverse = text[::-1]

if text == reverse:
    print("The given string is a Palindrome.")
else:
    print("The given string is not a Palindrome.")
```

OUTPUT:

```
DIYA S 25bai0235
Enter a string: aaaaabbbbaaaaa
The given string is a Palindrome.
```